

Title: ((Yin)) (title not final)

Subtitle: Online tool for beginning study of Mandarin tones.

Elevator Pitch:

There are many apps that teach Chinese, but few that teach about tone and fewer still that facilitate actual tone practice. Popular apps including Rosetta Stone, Duolingo, and ChineseSkill offer the ability for the user to record their speech and be graded on it, but the user is only given a pass/fail. There is no chance for the user to recover and learn from their mistake because they are only made aware they made a mistake without being told what the mistake is. Yin will provide graphs showing the pitch contour of user recordings and automatically compare them to ideal native speaker contours. This will help the user identify problem areas, and built in exercises help the user practice those areas.

Research questions:

1. What methodologies already exist for teaching Mandarin tones to native English speakers?
2. What apps currently exist to address tones?
 - a. What is their methodology?
 - b. What is successful about these designs?
 - c. What isn't successful?
3. What platform and technologies best facilitate reaching the target audience?
 - a. Are computer microphones of high enough quality to work with this?
4. Which tone evaluation style is most functional (e.g. template matching)
5. What are the biggest challenges in learning tone for native English speakers?
6. How could this tool be used by teachers?
 - a. By independent learners?
7. What UI design facilitates easy comprehension of tone?
8. How can the UX design and feedback mechanisms be used to encourage learners to continue?

Proposal: We will create a Chinese language learning web application that will take in audio from a user, identify and match tones against proper Chinese pronunciation, and guide the user toward improving their speech.

Feature List:

- An introduction to tones in Mandarin Chinese
 - How word meaning is differentiated by tone
 - The 4 tones
 - Tone Sandhi (some tone combinations cause the tones to change)
- Tone disentanglement
 - Play a recording of a tone, then another recording of a tone
 - Prompt user to compare if tones are the same or different

- Provide feedback
 - Tell the user immediately after their response if their response is correct
 - If not correct, provide a pitch contour for each tone
 - Replay each tone, highlighting the corresponding graph
 - Test in this fashion until the user passes 10 consecutive trails
- Tone identification
 - Recording of a tone is played
 - User is given a choice of 1, 2, 3, 4
 - User responds with the number they believe matches the tone in the recording
 - User is told immediately if they selected the correct answer
 - If not correct, provide a pitch contour for test tone while replaying test tone
 - Provide pitch contour for the tone of the incorrect guess, playing a recording of that tone
- Practice for each tone in isolation, and for each possible tone combination
 - Give example
 - Prompt user to mimic example
 - Record user input
 - Display pitch contour of user input
 - Evaluate user input
 - Compare user input against native speaker
 - Display pitch contour of native speaker
 - Highlight errors in user input
 - Provide insight for improvement